Grade 11 Chemistry Final Exam Revision Guide

Source basis: Public preview metadata and visible topic structure from the Scribd page "Grade 11 Chemistry Final Exam Guide | Gr 11 - Final Exam-Chemistry Rev WS". The Scribd page identifies the source document as a 12-page Grade 11 chemistry revision worksheet for the 2024-2025 final examination. This guide is newly written from the visible topics and does not reproduce the worksheet verbatim.

Exam Pattern From The Public Preview

- Total questions: 33 in the exam paper, with internal choice in some questions.
- Section A: 16 multiple-choice questions, 1 mark each.
- Section B: 5 very short answer questions, 2 marks each.
- Section C: 7 short answer questions, 3 marks each.
- Section D: 2 case-based questions, 4 marks each.
- Section E: 3 long answer questions, 5 marks each.
- Calculators and log tables are not allowed.

Main Topics To Revise

- Atomic structure: photoelectric effect, quantum theory, electronic configuration, orbital diagrams, ionisation enthalpy, electron gain enthalpy, and periodic trends.
- Periodic classification: blocks, periods, groups, shielding effect, effective nuclear charge, and anomalous trends.
- Chemical bonding: ionic character, hybridisation, molecular orbital theory, bond order, magnetic character, dipole moment, hydrogen bonding, and molecular shape.
- Thermodynamics: state functions, enthalpy, entropy, Gibbs energy, phase changes, adiabatic processes, bond energy, and spontaneity.
- Chemical equilibrium: Le Chatelier's principle, Kc, Kp, reverse reactions, and changes in equilibrium conditions.
- Organic chemistry: isomerism, alkenes, aromaticity, ozonolysis, electrophiles, nucleophiles, resonance, hyperconjugation, carbocation stability, Markovnikov addition, peroxide effect, and bond-line formulas.
- Mole concept: empirical formula, molecular formula, Avogadro's number, and combustion calculations.

Core Formula Sheet

- E = hv
- c = lambda v
- E = hc / lambda
- Energy per mole = energy per photon x Avogadro constant

- Delta G = Delta H - T Delta S
- At equilibrium for a phase change or reaction condition, Delta G = 0
- Kc = product concentration terms / reactant concentration terms
- Kp expression uses partial pressures for gases only
- For reverse reaction, K reverse = 1 / K forward
- Bond enthalpy change = energy to break bonds - energy released forming bonds
- Bond order = (bonding electrons - antibonding electrons) / 2
- Empirical formula gives the simplest whole-number atom ratio.

Quick Concept Notes

Atomic Structure And Periodicity

The photoelectric effect supports the particle nature of light because electrons are emitted only when incident radiation has enough frequency, regardless of intensity. Electronic configuration explains periodic placement and many chemical trends. Across a period, ionisation enthalpy generally increases because nuclear attraction increases while shielding does not rise enough to fully offset it.

Electron gain enthalpy becomes more negative when an atom accepts an electron easily. Noble gases have positive electron gain enthalpy because adding an electron would require a new shell or disturb a stable configuration. Fluorine is smaller than chlorine, so added-electron repulsion can make fluorine's electron gain enthalpy less negative than chlorine's.

Chemical Bonding

Greater electronegativity difference usually means greater ionic character. Hybridisation affects molecular shape and electronegativity: sp carbon has more s-character than sp2 carbon, and sp2 carbon has more s-character than sp3 carbon. Molecular orbital theory predicts whether a molecule can exist through bond order. A bond order of zero means the molecule is not stable.

Thermodynamics

State functions depend only on the state of a system, not the path taken. Enthalpy, entropy, internal energy, and Gibbs energy are state functions; heat and work are path functions. A process is spontaneous when Delta G is negative. For positive Delta H and positive Delta S, the process becomes spontaneous at high temperature when T Delta S exceeds Delta H.

Equilibrium

Le Chatelier's principle predicts the direction of shift when concentration, pressure, volume, or temperature changes. A catalyst does not change the value of K or the final equilibrium composition; it only helps the system reach equilibrium faster.

Organic Chemistry

Position isomers differ in the location of a functional group or multiple bond. Geometric isomerism occurs when restricted rotation creates different spatial arrangements. Carbocations are stabilized by electron-releasing groups, hyperconjugation, and resonance. In normal addition of HBr to propene, the major product follows Markovnikov's rule; in the presence of peroxide, anti-Markovnikov addition can occur.

Practice Section A: Multiple Choice

Choose the best answer.

1. The photoelectric effect provides evidence that light can behave as
    A. only a wave
    B. only a magnetic field
    C. particles called photons
    D. a stationary medium

2. A reaction with positive Delta H and positive Delta S is spontaneous when
    A. T Delta S is greater than Delta H
    B. T Delta S is less than Delta H
    C. Delta S equals zero
    D. temperature is always zero

3. Le Chatelier's principle predicts
    A. the exact value of Kc
    B. the direction an equilibrium shifts after stress
    C. the molecular mass of a compound
    D. the bond order of a molecule

4. Which is not a state function?
    A. enthalpy
    B. internal energy
    C. entropy
    D. work

5. The isomerism shown by propan-1-ol and propan-2-ol is
    A. chain isomerism
    B. position isomerism
    C. metamerism
    D. geometric isomerism

6. At dynamic equilibrium,
    A. both forward and reverse reactions stop
    B. only the forward reaction continues

C. forward and reverse rates are equal

D. product concentration must be zero

7. The enthalpy of sublimation equals

A. enthalpy of fusion plus enthalpy of vaporization

B. enthalpy of fusion only

C. enthalpy of vaporization only

D. twice the ionisation enthalpy

8. A nucleophile is usually a species that

A. accepts an electron pair

B. donates an electron pair

C. has no electrons

D. is always a noble gas

9. Which hybridisation normally gives a bond angle close to 120 degrees?

A. sp

B. sp2

C. sp3

D. dsp2

10. The first ionisation enthalpy generally increases across a period because

A. atomic radius increases greatly

B. effective nuclear charge increases

C. shielding by inner electrons becomes infinite

D. atoms become isotopes

11. The molecular formula C2H4 has empirical formula

A. C2H4

B. CH2

C. CH4

D. C4H8

12. Which particle arrangement explains isotopes having similar chemical behaviour?

A. same number of protons and electrons

B. same number of neutrons only

C. same mass number

D. same number of nuclei

13. An adiabatic process has

A. q = 0

B. w = 0 always

C. Delta H = 0 always

D. pressure = 0

14. For a stable diatomic molecule, molecular orbital theory requires
    A. bond order less than zero
    B. bond order equal to zero
    C. bond order greater than zero
    D. no bonding electrons

15. The major product of adding HBr to propene without peroxide is
    A. 1-bromopropane
    B. 2-bromopropane
    C. propane
    D. propene oxide

16. A catalyst changes
    A. the equilibrium constant
    B. the final equilibrium composition
    C. the rate at which equilibrium is reached
    D. the sign of Delta H

Practice Section B: Very Short Answer

17. Draw the orbital box diagram for oxygen, atomic number 8.

18. Explain why the valence configuration of copper is commonly written as 3d10 4s1 instead of 3d9 4s2.

19. Arrange C-H, N-H, O-H, and F-H in increasing ionic character.

20. Write the general outer electronic configuration of a Group 1 element and predict the formula of its oxide.

21. Define electron gain enthalpy in one or two sentences.

Practice Section C: Short Answer

22. Explain why fluorine has a less negative electron gain enthalpy than chlorine.

23. Calculate Kc for 2SO2(g) + O2(g) <=> 2SO3(g) when [SO2] = 0.60 M, [O2] = 0.82 M, and [SO3] = 1.90 M.

24. Explain the difference between electrophiles and nucleophiles with one example of each.

25. Radiation of wavelength 242 nm ionises sodium. Calculate the ionisation energy per mole using E = hc/lambda. Use h = 6.626 x 10^-34 J s, c = 3.00 x 10^8 m/s, and NA = 6.022 x 10^23 mol^-1.

26. Explain why NH_3 has a higher dipole moment than NF_3, even though fluorine is more electronegative than nitrogen.

27. Describe how ethyne can be converted to benzene. Include the reaction type.

28. Explain why alkyl groups stabilize carbocations.

Practice Section D: Case-Based Questions

29. Periodic table case: Elements are arranged by increasing atomic number, and the block of an element depends on the subshell receiving the last electron. Answer the following.

A. What block contains elements whose differentiating electron enters an f orbital?

B. Why do elements in the same group often show similar chemical properties?

C. Which has greater shielding ability in the same shell: s, p, d, or f electrons?

D. Why are lanthanoids and actinoids placed separately below the main periodic table?

30. Carbocation case: Carbocation stability depends on inductive effect, hyperconjugation, and resonance. Answer the following.

A. Arrange methyl, primary, secondary, and tertiary carbocations in increasing stability.

B. Why does resonance increase carbocation stability?

C. Why does a tertiary carbocation form more easily than a primary carbocation?

D. Give one reason allylic carbocations are unusually stable.

Practice Section E: Long Answer

31. Explain the addition of HBr to propene under normal conditions and under peroxide conditions. Write the major product in each case and identify the rule followed.

32. Draw or describe the molecular orbital filling of N_2. Calculate its bond order and predict whether it is diamagnetic or paramagnetic.

33. A compound contains 4.07 percent hydrogen, 24.27 percent carbon, and 71.65 percent chlorine by mass. Its molar mass is 98.96 g/mol. Determine its empirical formula and molecular formula.

34. Use bond energies to estimate ΔH for $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$. Bond energies: H-H = 435 kJ/mol, Br-Br = 192 kJ/mol, H-Br = 368 kJ/mol.

35. Explain the photoelectric effect and state two observations that could not be explained by classical wave theory.

Answer Key And Worked Guidance

1. C
2. A
3. B
4. D
5. B
6. C
7. A
8. B
9. B
10. B
11. B
12. A
13. A
14. C
15. B
16. C

17. Oxygen: 1s2 2s2 2p4. In orbital boxes, 1s and 2s are paired; 2p has one paired orbital and two singly occupied orbitals according to Hund's rule.

18. A completely filled 3d subshell gives extra stability, so copper is better represented as 3d10 4s1.

19. C-H < N-H < O-H < F-H, because electronegativity difference with hydrogen increases across the period.

20. Group 1 outer configuration: ns1. Oxide formula: M2O.

21. Electron gain enthalpy is the enthalpy change when an electron is added to a neutral gaseous atom to form a gaseous anion.

22. Fluorine is very small, so the incoming electron experiences strong electron-electron repulsion in a compact 2p orbital. Chlorine is larger, so the added electron is accommodated with less repulsion.

23. Kc = [SO3]^2 / ([SO2]^2[O2]) = (1.90)^2 / ((0.60)^2(0.82)) = 12.2 approximately.

24. Electrophiles accept electron pairs, such as H+ or BF3. Nucleophiles donate electron pairs, such as OH- or NH3.

25. E per photon = hc/lambda = (6.626 x 10^-34)(3.00 x 10^8)/(242 x 10^-9) = 8.21 x 10^-19 J. Per mole = 8.21 x 10^-19 x 6.022 x 10^23 = 4.94 x 10^5 J/mol = 494 kJ/mol.

26. In NH3, the bond dipoles and lone-pair direction reinforce each other. In NF3, the N-F bond dipoles

oppose the lone-pair contribution, reducing the net dipole moment.

27. Ethyne can be converted to benzene by cyclic polymerisation: three molecules of ethyne combine to form benzene under suitable heat and catalyst conditions.

28. Alkyl groups release electron density through the positive inductive effect and hyperconjugation, reducing the electron deficiency at the positively charged carbon.

29. A. f-block. B. Same valence electron arrangement. C. s > p > d > f shielding. D. To keep the periodic table compact while preserving periodic relationships.

30. A. methyl < primary < secondary < tertiary. B. Resonance delocalizes positive charge. C. More alkyl groups donate electron density and allow more hyperconjugation. D. Allylic carbocations are resonance stabilized.

31. Normal HBr addition to propene gives 2-bromopropane by Markovnikov's rule through the more stable secondary carbocation. With peroxide, radical addition gives 1-bromopropane by anti-Markovnikov orientation.

32. N2 has 10 valence electrons. Filling the molecular orbitals gives bond order 3. All electrons are paired, so N2 is diamagnetic.

33. Assume 100 g: H = 4.07/1.008 = 4.04 mol, C = 24.27/12.01 = 2.02 mol, Cl = 71.65/35.45 = 2.02 mol. Divide by 2.02 to get C1H2Cl1, empirical formula CH2Cl. Empirical mass is 49.48 g/mol. Molar mass 98.96 g/mol is double, so molecular formula is C2H4Cl2.

34. Bonds broken = 435 + 192 = 627 kJ/mol. Bonds formed = 2(368) = 736 kJ/mol. Delta H = 627 - 736 = -109 kJ/mol.

35. The photoelectric effect is emission of electrons from a metal surface when light of sufficient frequency falls on it. Classical wave theory could not explain threshold frequency or the immediate emission of electrons once threshold frequency is met. Quantum theory explains light as photons with energy E = hv.

Final Revision Strategy

- Memorize definitions, but practise applying them to unfamiliar examples.
- For numerical questions, write the formula first and keep units visible.
- For organic mechanisms, identify the intermediate and explain stability.
- For equilibrium, state the stress, predicted shift, and reason.
- For molecular orbital questions, finish with bond order and magnetic character.